

Indian Healthcare Patient Records Analysis

Healthcare Data Analytics Project

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Tools Used: Python | Pandas | NumPy | Matplotlib | Power BI | Excel

Dataset Size: 100,000+ Patient Records

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1. Project Overview

This project focuses on analyzing the **Indian Healthcare Patient Records** dataset containing **100,000 patient records**. The objective is to understand patient demographics,

disease patterns, treatment outcomes, hospital performance, insurance coverage, and key health indicators using data analytics techniques. The project will follow a structured data analytics workflow, including data understanding, data cleaning, exploratory data analysis (EDA), feature engineering, and dashboard development to generate meaningful business insights that can support data-driven healthcare decisions.

Item	Details
Project Name	Indian Healthcare Patient Records Analysis
Domain	Healthcare Analytics
Dataset Source	Kaggle
Total Records	100,000
Total Columns	19
Primary Tool	Python (Kaggle Notebook)
Visualization Tool	Power BI
Database Tool	SQL
Libraries Used	Pandas, NumPy, Matplotlib

2. Business Problem Statement

Healthcare institutions collect large volumes of patient data, including demographic information, diagnoses, laboratory test results, treatment details, and hospital records. However, without proper analysis, this data remains underutilized, making it difficult to identify disease patterns, evaluate treatment effectiveness, compare hospital performance, understand patient health trends, and optimize healthcare resources.

The objective of this project is to analyze healthcare patient records and transform raw data into meaningful insights. These insights can help healthcare professionals and decision-makers improve patient care, evaluate treatment strategies, optimize hospital resources, and support data-driven decision-making.

3. Project Objectives

The primary objectives of this project are:

- Analyze patient demographics and healthcare trends.

- Identify the most common diseases and symptoms.
- Analyze regional variations in disease occurrence.
- Study key health indicators such as Blood Glucose, HbA1c, Total Cholesterol, and BMI.
- Evaluate treatment effectiveness across different treatment types.
- Compare patient outcomes between Government and Private hospitals.
- Analyze the impact of insurance coverage on treatment outcomes.
- Generate meaningful business insights using exploratory data analysis (EDA).
- Develop an interactive Power BI dashboard for healthcare decision-making.

4. Expected Business Value

The findings of this analysis can provide valuable support to healthcare organizations by:

- Identifying the most common diseases and patient health trends.
- Improving treatment planning through outcome analysis.
- Supporting hospital management in resource allocation.
- Understanding the impact of insurance coverage on patient outcomes.
- Monitoring key health indicators across different patient groups.
- Enabling faster and more informed decision-making through interactive dashboards.

5. Data Understanding

5.1 Dataset Overview

The **Indian Healthcare Patient Records** dataset contains **100,000 patient records** and **19 variables** representing different aspects of patient healthcare information. The dataset includes demographic details, medical diagnoses, laboratory test results, treatment information, imaging records, hospital type, insurance status, and Body Mass Index (BMI).

The dataset consists of a combination of numerical, categorical, and boolean variables, making it suitable for descriptive analysis, comparative analysis, and healthcare business intelligence.

5.2 Variable Types

The dataset contains a combination of numerical, categorical, and boolean variables. The `df.info()` function was used to understand the data structure, identify data types, detect missing values, and estimate memory usage.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 19 columns):
 #   Column                                  Non-Null Count  Dtype  
---  -
 0   Patient_ID                             100000 non-null object  
 1   Age                                     100000 non-null int64   
 2   Gender                                 100000 non-null object  
 3   Region                                 100000 non-null object  
 4   Socioeconomic_Status                   100000 non-null object  
 5   Occupation                             100000 non-null object  
 6   Visit_Date                             100000 non-null object  
 7   Symptoms                               100000 non-null object  
 8   Primary_Diagnosis                       100000 non-null object  
 9   Blood_Glucose_mg_dL                    100000 non-null float64  
10  HbA1c_%                                100000 non-null float64  
11  Total_Cholesterol_mg_dL                 100000 non-null float64  
12  Treatment_Type                          100000 non-null object  
13  Treatment_Outcome                       100000 non-null object  
14  Imaging_Type                            64969 non-null  object  
15  Imaging_Findings                        100000 non-null object  
16  Hospital_Type                           100000 non-null object  
17  Insurance_Covered                       100000 non-null bool      
18  BMI                                     100000 non-null float64  
dtypes: bool(1), float64(4), int64(1), object(13)
memory usage: 13.8+ MB
```

Dataset information generated using `df.info()` showing the dataset structure, data types, non-null values, and memory usage.

Data Type	Number of Columns	Columns
Numerical	5	Age, Blood_Glucose_mg_dL, HbA1c_%, Total_Cholesterol_mg_dL, BMI
Categorical (Object)	13	Patient_ID, Gender, Region, Socioeconomic_Status, Occupation, Visit_Date, Symptoms, Primary_Diagnosis, Treatment_Type, Treatment_Outcome, Imaging_Type, Imaging_Findings, Hospital_Type
Boolean	1	Insurance_Covered

Key Observations

- Dataset contains 100,000 records.
- Dataset has 19 columns.
- Visit_Date is stored as an object.

- Imaging_Type contains missing values.
- Memory usage is 13.8 MB.

5.3 Summary Statistics

A summary statistical analysis was performed on all numerical variables to understand the overall distribution and identify potential data quality issues before beginning the data cleaning process.

Key Observations

- The dataset contains **100,000 complete records** for all numerical variables, indicating that there are **no missing values** in the numerical columns.
- The average patient age is **44.45 years**, suggesting that the dataset mainly represents the adult population.
- The average **BMI** is **24.56**, which is close to the overweight threshold.
- The average **Blood Glucose** level is **127.70 mg/dL**.
- The average **HbA1c** level is **6.40%**.
- The average **Total Cholesterol** level is **216.14 mg/dL**.
- A **negative Blood Glucose value (-11.7 mg/dL)** was identified, which is medically impossible and indicates a potential data quality issue that will be addressed during the data cleaning phase.
- The maximum values of Age, BMI, HbA1c, and Total Cholesterol are medically possible and may represent high-risk patients.

5.4 Missing Value Analysis

A missing value analysis was performed to identify incomplete records within the dataset.

Column	Missing Values	Missing Percentage	Observation
Imaging_Type	35,031	35.03%	Missing values are likely associated with patients who did not undergo any imaging procedure.

5.5 Initial Data Quality Observations

The initial review of the dataset identified the following observations:

Observation	Status
Dataset successfully loaded	
Numerical columns contain no missing values	
Imaging_Type contains missing values	
Blood_Glucose_mg_dL contains an invalid negative value	
Visit_Date is currently stored as an object data type and requires conversion to datetime for time-based analysis	
Dataset is ready for the Data Cleaning phase	

6. Data Cleaning

6.1 Duplicate Record Analysis

Purpose

Duplicate records can lead to incorrect analysis, inaccurate patient counts, and misleading business insights. Therefore, duplicate records were checked before performing further data cleaning and analysis.

Method

The `uplicated().sum()` function in Pandas was used to identify completely duplicated records in the dataset.

Result

No duplicate records were found in the dataset.

Business Observation

The dataset contains **100,000 unique records**, indicating good data quality with respect to duplicate entries. Therefore, no duplicate records were removed during the data cleaning process.

6.2 Missing Value Treatment

Purpose

Missing values can affect the accuracy of data analysis and visualizations. Therefore, missing values were carefully analyzed before applying any treatment.

Method

The **Imaging_Type** column contained **35,031 missing values**. These records were compared with the **Imaging_Findings** column, where the corresponding values were **"No imaging performed"**. This confirmed that the missing values represented patients who did not undergo any imaging procedure rather than data entry errors.

Action Taken

The missing values in the **Imaging_Type** column were replaced with **"No Imaging"** to preserve the business meaning and ensure consistency during analysis.

```
df[df['Imaging_Type'].isnull()][['Imaging_Type', 'Imaging_Findings']].head(10)
```

	Imaging_Type	Imaging_Findings
1	NaN	No imaging performed
2	NaN	No imaging performed
6	NaN	No imaging performed
10	NaN	No imaging performed
11	NaN	No imaging performed
12	NaN	No imaging performed
15	NaN	No imaging performed
17	NaN	No imaging performed
18	NaN	No imaging performed
21	NaN	No imaging performed

Figure 6.2: Validation showing that missing values in the **Imaging_Type** column correspond to patients for whom no imaging procedure was performed.

Result

After treatment, the dataset contained **no missing values**.

6.3 Invalid Value Analysis

Purpose

To identify medically impossible values that may affect the accuracy of analysis.

Observation

A total of **6 records (0.006%)** contained negative Blood Glucose values. Since Blood Glucose cannot be negative, these values were identified as invalid.

Decision

The invalid Blood Glucose values were converted to missing values (**NaN**) instead of deleting the complete records.

Reason

Only the Blood Glucose values were invalid, while all other patient information remained valid. Converting these values to **NaN** preserved the remaining patient information for further analysis and allowed appropriate imputation in the next step.

6.4 Missing Value Imputation (Blood Glucose)

Purpose

To handle invalid Blood Glucose values while preserving patient records.

Method

The invalid negative Blood Glucose values were first converted into missing values (**NaN**). Since the distribution contains high glucose values that influence the average, the **median** was selected as the imputation method because it is less affected by extreme values.

Action Taken

The missing Blood Glucose values were replaced using the **median value (118.9 mg/dL)**.

Result

All missing values in the **Blood_Glucose_mg_dL** column were successfully imputed, and the dataset became complete for further analysis.

6.5 Data Cleaning Summary

The data cleaning process was performed to improve the quality, consistency, and reliability of the dataset before conducting exploratory data analysis (EDA). During this phase, duplicate records, missing values, invalid values, and data types were carefully examined and treated using appropriate data cleaning techniques. All cleaning decisions were based on business logic and healthcare domain understanding to ensure that valuable patient information was preserved.

Data Cleaning Log

Data Quality Issue	Observation	Action Taken	Business Reason
Duplicate Records	No duplicate records found	No action required	Dataset already contained unique records.
Visit_Date Data Type	Stored as <code>object</code>	Converted to <code>datetime</code>	Enables time-based analysis such as monthly and yearly trends.
Missing Values (<code>Imaging_Type</code>)	35,031 missing values	Replaced with "No Imaging"	Missing values represented patients who did not undergo any imaging procedure.
Invalid Blood Glucose Values	6 negative values	Converted to <code>NaN</code> and imputed using the median (118.9 mg/dL)	Negative glucose values are medically impossible. Median preserves the overall distribution and is less affected by extreme values.

7. Exploratory Data Analysis (EDA)

Introduction

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the healthcare dataset, identify important trends, detect patterns, and generate business

insights. Various statistical summaries and visualizations were used to analyze patient demographics, disease distribution, treatment outcomes, hospital performance, insurance coverage, and health indicators.

7.1 Patient Demographic Analysis

Patient demographic analysis was performed to understand the distribution and characteristics of patients based on age, gender, and geographical region. These demographic factors help identify the major patient groups and regional healthcare demand patterns.

7.1.1 Age Distribution of Patients

Purpose:

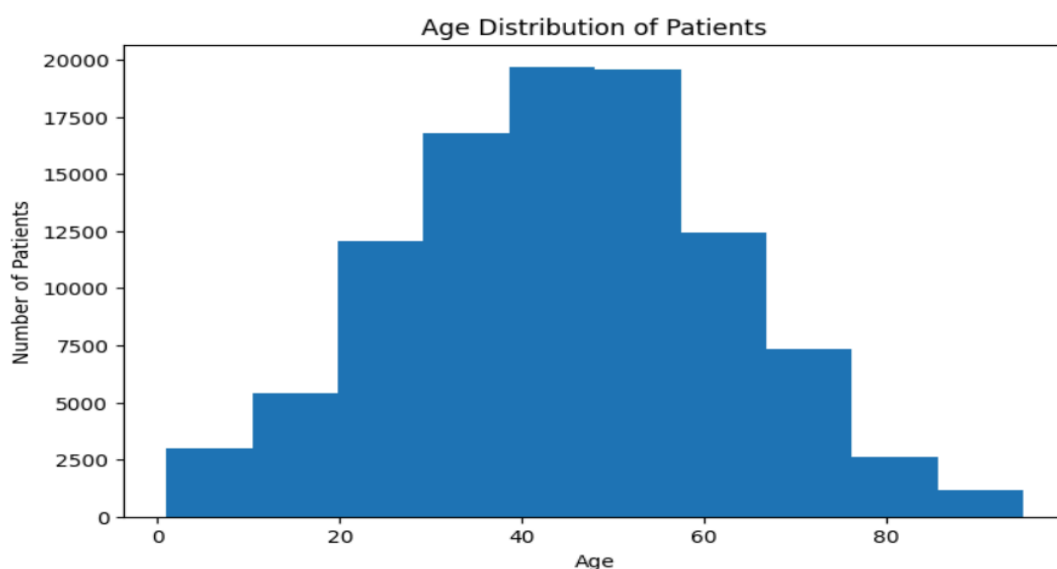
To analyze the distribution of patients across different age groups.

Visualization:

A histogram was used to understand the distribution of patient ages.

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.hist(df["Age"], bins=10)
plt.title("Age Distribution of Patients")
plt.xlabel("Age")
plt.ylabel("Number of Patients")
plt.show()
```



The dataset includes patients across a wide range of age groups, with a higher concentration of patients in the adult and middle-aged population.

Business Insight:

Understanding the age distribution can help healthcare organizations plan age-specific healthcare services and allocate medical resources according to the healthcare needs of different age groups.

7.1.2 Gender Distribution

Purpose:

To analyze the distribution of patients based on gender.

Visualization:

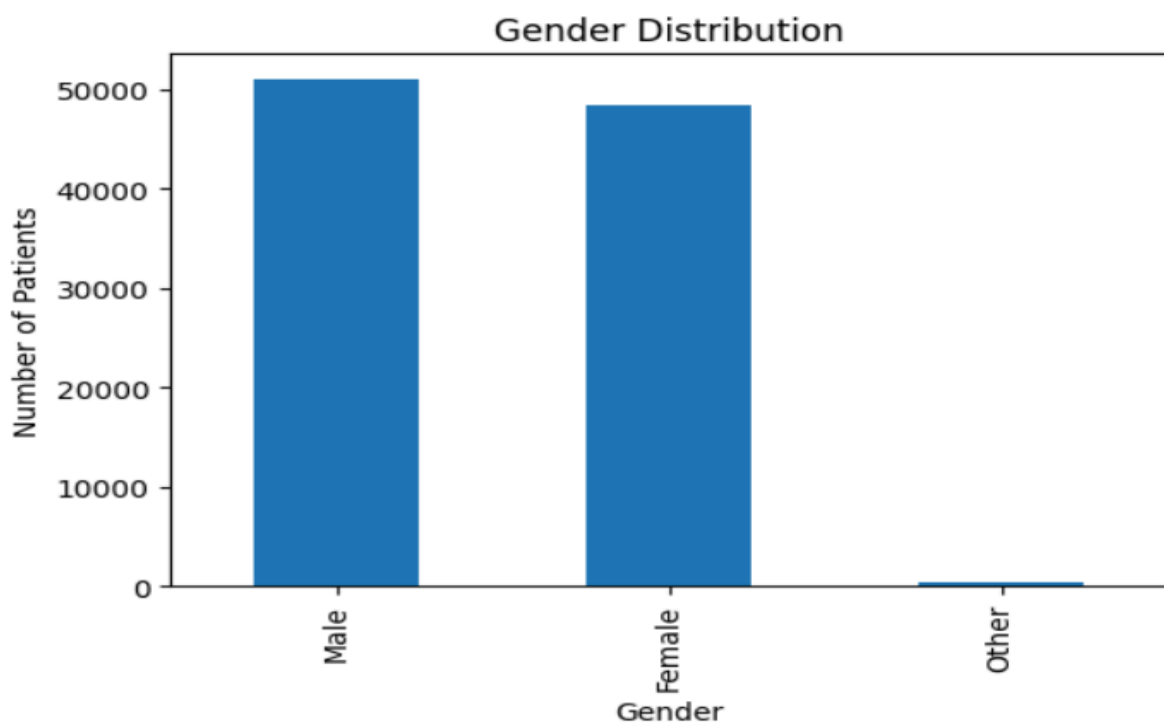
A bar chart was used to compare the number of patients across different gender categories.

Key Observation:

The chart shows the distribution of patients across the available gender categories in the dataset.

Business Insight:

Gender-based analysis can help healthcare organizations identify differences in patient demographics and support more targeted healthcare planning and service delivery.



7.1.3 Patients by Region

Purpose:

To analyze the geographical distribution of patients across different regions.

Visualization:

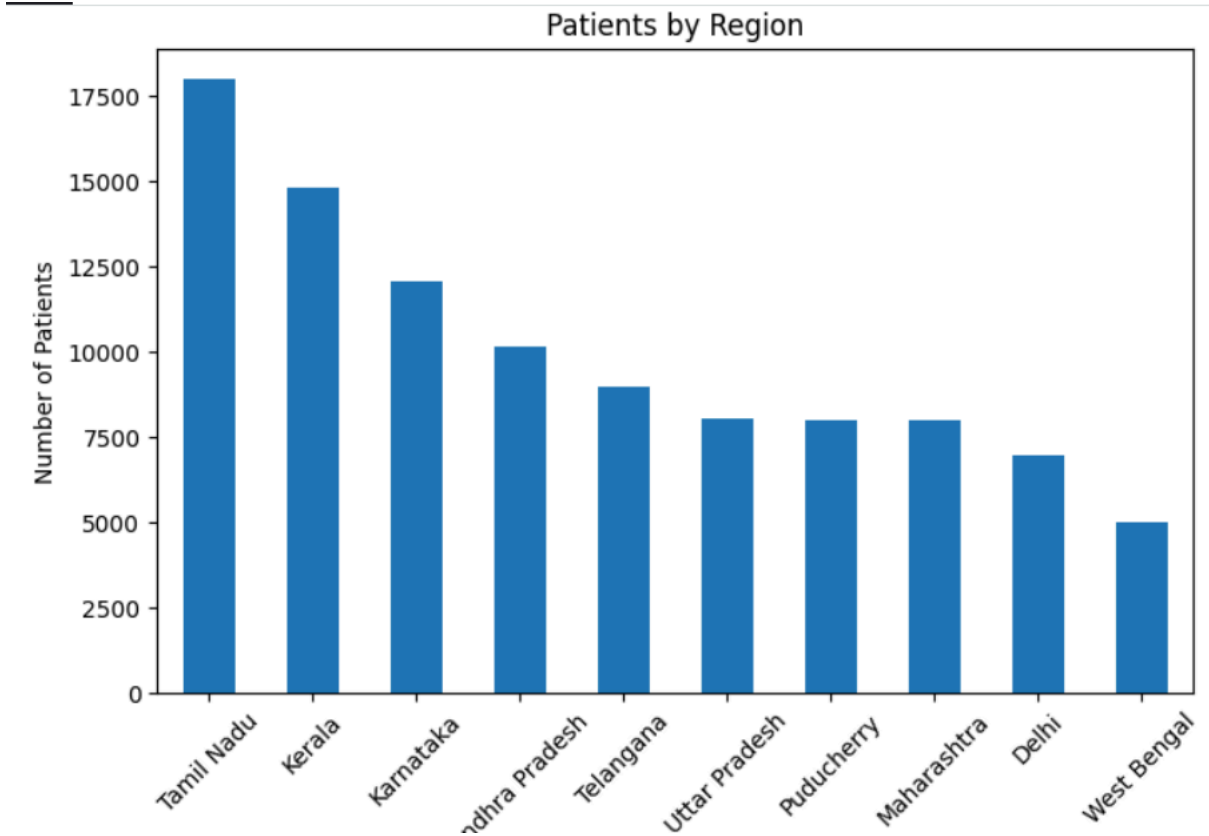
A bar chart was used to compare the number of patients across different regions.

Key Observation:

The number of patients varies across different regions, indicating differences in patient representation and healthcare demand.

Business Insight:

Regional patient distribution can help healthcare organizations identify areas with higher patient demand and support better planning of healthcare resources, facilities, and services.



7.2 Disease Analysis

Disease analysis was performed to identify the most common primary diagnoses among patients and understand how disease patterns vary across different age groups. This analysis helps healthcare organizations identify major disease burdens and understand the healthcare needs of different patient populations.

7.2.1 Primary Diagnosis Distribution

Purpose:

To identify the most frequently occurring diseases among the patients in the dataset.

Visualization:

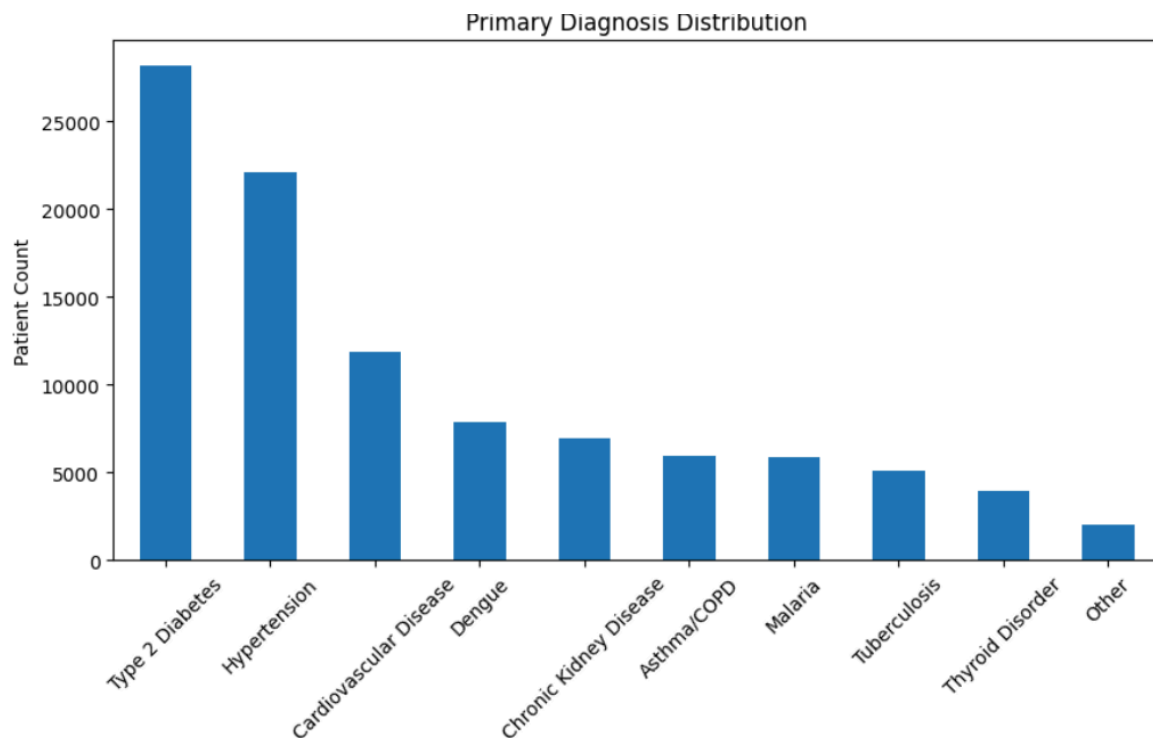
A bar chart was used to compare the number of patients across different primary diagnosis categories.

Key Observation:

The distribution of patients varies across different primary diagnoses, with some diseases occurring more frequently than others.

Business Insight:

Identifying the most common diseases can help healthcare organizations prioritize medical resources, treatment facilities, and disease-specific healthcare programs according to patient demand.



7.2.2 Disease Distribution by Age Group

Purpose:

To analyze the distribution of different diseases across various age groups.

Visualization:

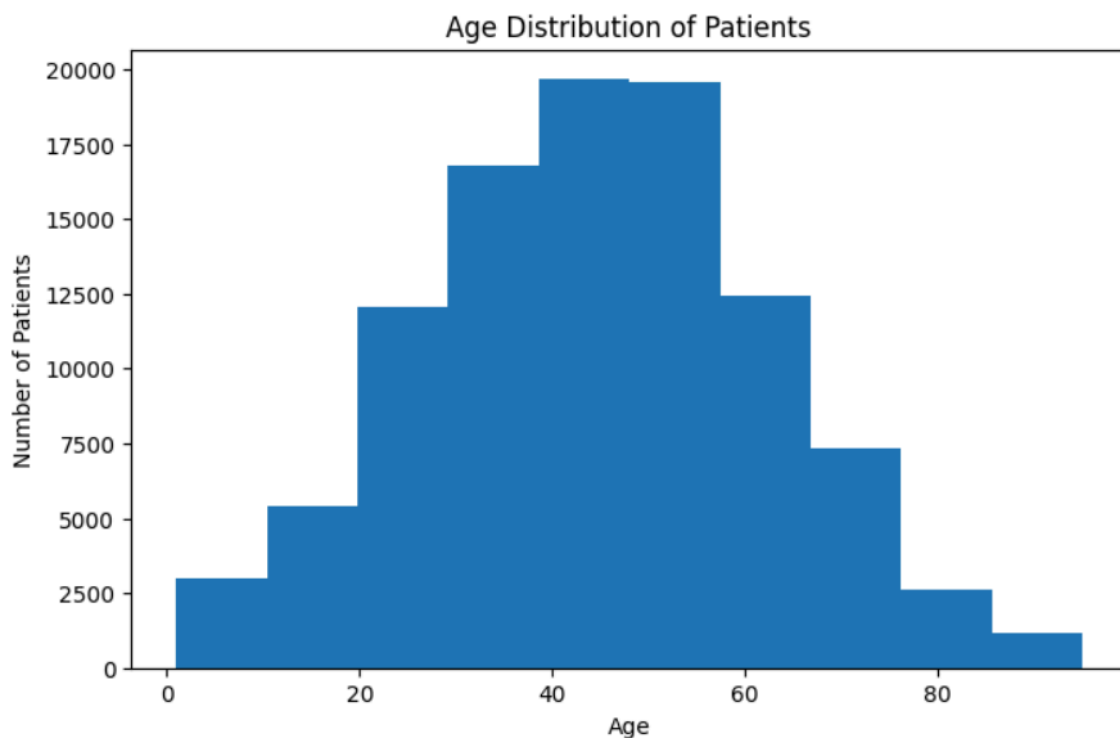
A grouped bar chart or stacked bar chart can be used to compare disease patterns across different age groups.

Key Observation:

The occurrence of diseases varies across different age groups, indicating that certain health conditions may be more common among specific patient age groups.

Business Insight:

Age-based disease analysis can help healthcare providers develop targeted prevention programs, screening strategies, and treatment plans for different age groups.



7.3 Health Indicator Analysis

Healthcare indicators were analyzed to understand the overall health characteristics of patients and identify variations in important clinical measurements. The analysis focused on Blood Glucose, HbA1c, Total Cholesterol, and BMI, as these indicators provide important information about patients' metabolic and general health conditions.

7.3.1 Blood Glucose Distribution

Purpose:

To understand the distribution of Blood Glucose levels among patients.

Visualization:

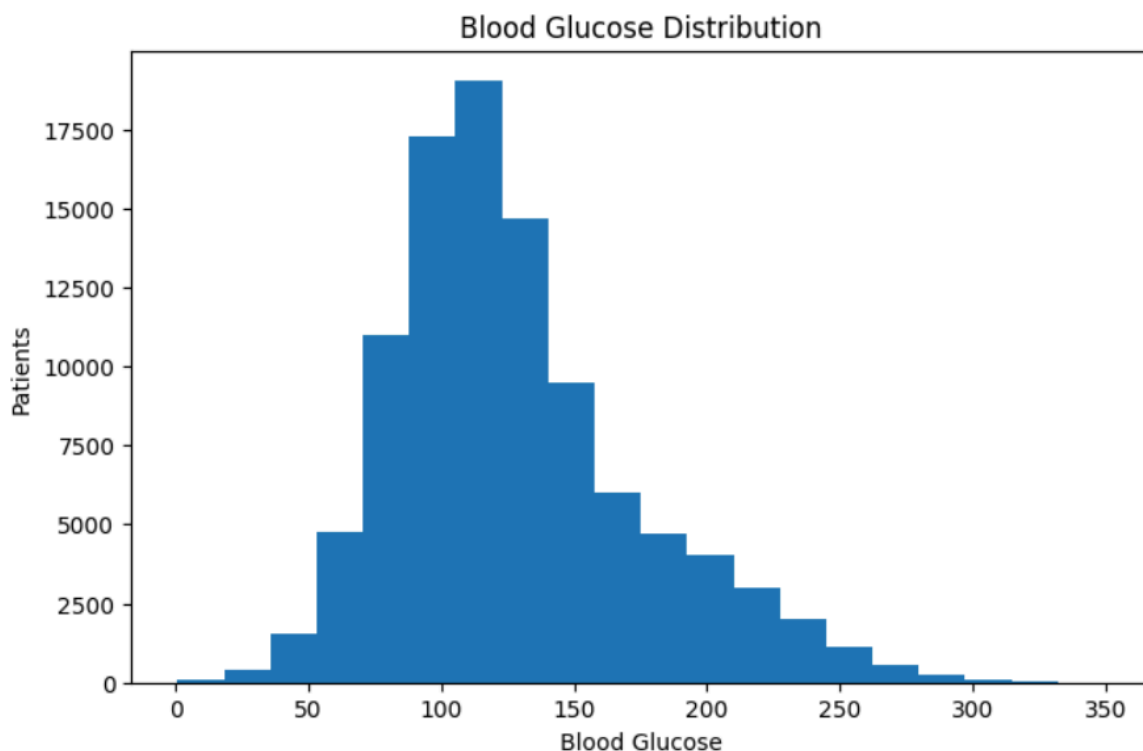
A histogram was used to analyze the distribution of Blood Glucose values across the patient population.

Key Observation:

The distribution shows variation in Blood Glucose levels among patients, with most observations concentrated within a particular range and some patients having relatively higher glucose levels.

Business Insight:

Monitoring Blood Glucose levels can help healthcare organizations identify patients who may require additional monitoring and support for glucose-related health risks.



8. Power BI Dashboard

An interactive healthcare dashboard was developed using **Microsoft Power BI** to analyze 100,000 patient records and provide a comprehensive overview of patient demographics, diseases, treatment patterns, and healthcare trends.

The dashboard includes key performance indicators such as:

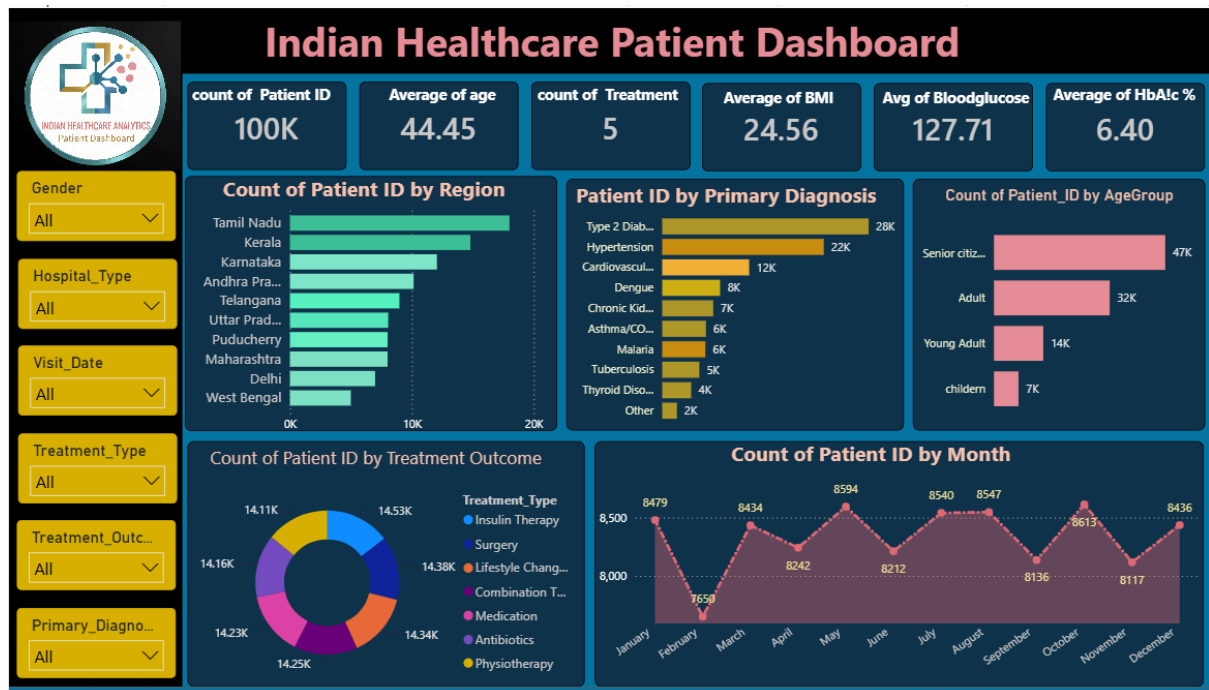
- **Total Patients:** 100K
- **Average Patient Age:** 44.45 years
- **Average BMI:** 24.56
- **Average Blood Glucose:** 127.71
- **Average HbA1c:** 6.40%

The dashboard provides interactive filters for **Gender, Hospital Type, Visit Date, Treatment Type, Treatment Outcome, and Primary Diagnosis**, allowing users to explore the data from different perspectives.

Key Visualizations:

- **Region-wise Patient Distribution:** Shows the number of patients across different Indian regions.
- **Primary Diagnosis Analysis:** Highlights the most common diagnoses, including Type 2 Diabetes, Hypertension, Cardiovascular Diseases, and Dengue.
- **Age Group Distribution:** Compares patients across Children, Young Adults, Adults, and Senior Citizens.
- **Treatment Type Distribution:** Shows the distribution of patients across different treatment types such as Insulin Therapy, Surgery, Medication, Antibiotics, and Physiotherapy.
- **Monthly Patient Trend:** Displays the monthly variation in patient visits throughout the year.

The dashboard enables users to interactively filter and analyze healthcare data, making complex information easier to understand and supporting data-driven decision-making.



9. Key Insights

Tumhare dashboard ke according:

- The dataset contains **100K patient records**, providing a large-scale overview of healthcare patterns.
- **Senior Citizens** represent the largest age group, with approximately **47K patients**.
- **Type 2 Diabetes** is the most common primary diagnosis, followed by **Hypertension**.
- **Tamil Nadu** has the highest number of patients among the displayed regions.
- The average **Blood Glucose level is 127.71**, while the average **HbA1c is 6.40%**.
- The average **BMI is 24.56**, providing an overview of the general weight-related health profile of patients.
- Monthly patient counts show variation throughout the year, with the highest patient count observed in **May**.
- Treatment types are relatively distributed across different categories, indicating varied treatment requirements among patients.

10. Conclusion

This project successfully analyzed 100,000 Indian healthcare patient records to identify important patterns in patient demographics, diseases, health indicators, treatment types, and monthly healthcare trends.

The analysis revealed that senior citizens formed the largest patient group, while Type 2 Diabetes and Hypertension were among the most common diagnoses. Regional analysis also highlighted differences in patient distribution across various Indian regions. Key health indicators such as Blood Glucose, HbA1c, and BMI provided additional insights into the overall health profile of the patients.

The interactive Power BI dashboard transformed complex healthcare data into meaningful visual insights. With interactive filters and key performance indicators, users can easily explore patient trends based on gender, hospital type, visit date, treatment type, treatment outcome, and diagnosis.

Overall, this project demonstrates the complete data analysis process, from data cleaning and exploratory analysis to dashboard development and insight generation. The findings can help healthcare stakeholders better understand patient patterns, identify major health concerns, and support more informed, data-driven decision-making.